

EduGenie

explanation_module.py — Concept Explanation

Concept explanation module. The project document specifies LaMini-Flan-T5 as the lightweight model for simplifying complex topics into readable explanations.

Complete Python Code

```
from transformers import pipeline

MODEL_NAME = "MBZUAI/LaMini-Flan-T5-783M"
_generator = None

def _get_generator():
    global _generator
    if _generator is None:
        _generator = pipeline("text2text-generation", model=MODEL_NAME)
    return _generator

def explain_topic(topic: str) -> str:
    prompt = (
        "Explain this topic in simple language for a student. "
        "Give a definition, important points and one example. "
        f"Topic: {topic}"
    )
    result = _get_generator()(prompt, max_new_tokens=350, do_sample=False)
    return result[0]["generated_text"]
```